

RIAD MAHAMUD PIASH

Mechanical Engineer | Digital Twin & Intelligent Systems

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Location: Gazipur, Bangladesh

EDUCATION

Dhaka University of Engineering and Technology (DUET)
Bachelor of Science (B.Sc.) in Mechanical Engineering
CGPA: 3.34/4.00

GAZIPUR, BANGLADESH
Oct 2021 - Oct 2025

BCMC College of Engineering & Technology
Diploma in Marine Technology
CGPA: 3.86/4.00 (Merit Position 2 / 70)

JASHORE, BANGLADESH
July 2016 – Feb 2021

TEACHING & MENTORING EXPERIENCE

- **Lecturer (Part-Time), Department of Marine Technology** BCMC, BANGLADESH
Course: Mechanical Engineering Drawing, Computer-Aided Design (CAD),
Computer Programming Language Sessional (Python, Matlab) Mar 2020 – July 2021
- **Lecturer, Department of Mechanical Engineering** NEXUS, BANGLADESH
Course: Mechanics, Thermodynamics, Fluid Mechanics, Drawing Dec 2021-Dec 2022

PROJECTS

- I. **Cozmo Clance** Techfest IIT Bombay, Bangladesh Zonal
Position: 1st Runners-up October 25, 2024
- II. **Meshmerise Robot** Techfest IIT Bombay, Bangladesh Zonal
Position: 4th place October 25, 2024
- III. **Autonomous Trash Collector-Robot** Dhaka University of Engineering & Technology, Bangladesh
Capability: Autonomous waste collection, classification, disposal, obstacle avoidance, and return-to-base.
February 12, 2025
- IV. **Agro-Robot** Dhaka University of Engineering & Technology, Bangladesh
Capability: Autonomous Disease Detection & Self-Remedy Systems April 12, 2025
- V. **Voice Braille Navigator (VBN)** City University, Dhaka, Bangladesh
Capability: Obstacle detection, location guidance, voice-based browsing with audio reading, and Braille keyboard control January 02, 2026

RESEARCH

I. Physics-Based Digital Twin Framework for Real-Time Thermal State Estimation of Lithium-Ion Batteries Using Extended Kalman Filter (Conference Paper)

Status: Accepted by IEEE International Conference on Engineering and Frontier Technologies 2026 (ICEFronT) Presentation ongoing.

- Developed physics-based battery digital twin model
- Implemented Extended Kalman Filter for internal temperature estimation
- Designed real-time monitoring framework for battery systems

II. Tomato and Potato Leaf Disease Classification Using YOLO V11 (Conference Paper)

Status: Accepted for Publication by API Publication.

III. Autonomous trash-collector robot → "Smart e-waste micro-EV"

Status: Manuscript prepared for submission to [IEEE]

IV. Autonomous disease-detection and remedies vehicle → "Agri e-scout"

Status: Manuscript review for submission to [IEEE]

V. Current Progresses in Hybrid Coating on Biodegradable Mg Alloys for Implant Applications (Review Paper)

Status: Submitted to **Journal of Coatings Technology and Research (JCTR)** (Under Review)

VI. A novel route for low-cost, scalable synthesis of phosphorus-doped graphene via electrochemical exfoliation with tunable bandgap

Status: Submitted to Royal Society of Chemistry (Under Review)

RESEARCH INTERESTS

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|----------------------------|---------------------------|-----------------------------|
| ❖ Digital Twin Systems | ❖ Physics-Based Modeling | ❖ AI for Autonomous Systems |
| ❖ Battery Thermal Modeling | ❖ Computational Materials | ❖ MATLAB-Based Modeling |

EXTRA-CURRICULAR ACTIVITIES

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|--|------------------|
| ➤ Executive member at DUET Robotics Club | DUET, Bangladesh |
| ➤ Executive member at DUET English Language Club | DUET, Bangladesh |
| ➤ VICE PRESIDENT at KAPOTAKKHO Student Welfare Association | DUET, Bangladesh |
| ➤ Executive member at DUET Society of Mechanical Engineering | DUET, Bangladesh |
| ➤ Member The American Society of Mechanical Engineers (ASME) | |

TECHNICAL SKILLS

Programming	: MATLAB, Python
Machine Learning	: PyTorch, YOLO
Modeling & Simulation:	MATLAB Modeling, ANSYS (CFD), Physics-Based System Modeling
Digital Twin &	Real-Time System Modeling, State Estimation (Extended Kalman
Autonomous Systems	: Filter)
Engineering Tools	: CATIA V5, SolidWorks, AutoCAD
Scientific Tools	: LaTeX, MS Office

RELEVANT COURSEWORK

➤ Deep Learning for Images with PyTorch	DataCamp, Nov 2024
➤ Image Processing in Python	DataCamp, Nov 2024
➤ Intermediate Python	DataCamp, Nov 2024
➤ Introduction to Python	DataCamp, Nov 2024
➤ Understanding Machine Learning	DataCamp, Nov 2024
➤ Python Data Analysis Course for Beginners	Udemy, Dec 2024

ADDITIONAL COURSE

Industrial Attachment Training	Bangladesh Road Transport Corporation (BRTC), Bangladesh
Subject: Vehicle Maintenance and Repair Training	Mar 2025 – April 2025
Industrial Internship	National Engineering Development Limited, Bangladesh
Subject: Programmable logic controller	Feb 2020 – May 2020
Training	Crist Church Trade School (CCTS), Jashore, Bangladesh
Subject: IC Engine Overhauling	July 2018 – Dec 2018
Training	Crist Church Trade School (CCTS), Jashore, Bangladesh
Subject: Marine Engine Overhauling	Jan 2019 – Mar 2019

WORKSHOP & VISITING

Workshop on Automotive Design	Dhaka University of Engineering and Technology (DUET), Bangladesh
Speaker: M. Ahsanul Karim, Thermal Engineer	Ford Motor Company, USA
Subject: Fluid Dynamics and CFD Applications	12 July 2025 – 15 July 2025
ONE DAY INDUSTRIAL VISIT	multibrand WORKSHOP Ltd, Tejgaon, Dhaka Bangladesh
Subject: Hybrid and Electric Car Observation	July 2025
ONE DAY INDUSTRIAL VISIT	Ashuganj Power Station Company Limited – APSCL, Bangladesh
ONE WEEK INDUSTRIAL VISIT	Khulna Shipyard Ltd, Khulna, Bangladesh

REFERENCES

Dr. Md. Mostaqur Rahman

Professor, Ph.D.

Dhaka University of Engineering and Technology (DUET), Gazipur, Bangladesh

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Dr. Nayeem Md. Lutful Huq

Professor, Ph.D.

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